

VEX Robotics & Base Bot Quiz

OCN NI Level 2 IT | Presentation #141

Topic: Build, drive, program and diagnose a VEX Base Bot

Microsoft Forms Quick Import tip: upload this PDF, choose Quiz, then review. Questions are numbered. Options use A. B. C. D. Each choice question ends with Answer: X so Forms can mark the correct option. Confirm points and feedback after import.

1. What is the main controller on a VEX Base Bot?

- A. V5 Brain
- B. Wireless remote only
- C. Arduino microcontroller
- D. Raspberry Pi

Answer: A

Points: 1

2. Which environment is used to program VEX robots?

- A. Only block-based coding
- B. VEXcode with Blocks or C++ options
- C. Python exclusively
- D. Hardware assembly only

Answer: B

Points: 1

3. What do rotation sensors (encoders) measure on the Base Bot?

- A. How far a wheel or motor has turned
- B. The temperature of motors
- C. The wireless signal strength
- D. Battery voltage only

Answer: A

Points: 1

4. Both drive sides are set to 50% but the bot pulls left. What should you check first?

- A. Rewrite the entire autonomous routine
- B. Left motor cable, port, and wheel bind
- C. Buy a new V5 Brain immediately
- D. Disable all sensors

Answer: B

Points: 1

5. Autonomous stops early every run. The distance sensor faces the floor. Likely cause?

- A. The battery is empty
- B. The stop condition is reading the floor
- C. VEXcode cannot run autonomous

D. The radio must be unplugged

Answer: B

Points: 1

6. Why should the Brain and battery be centred on the chassis?

- A. It looks neater in photos
- B. A lower, centred mass improves stability
- C. Sensors only work at the rear
- D. Wireless range doubles

Answer: B

Points: 1

7. In the Base Bot control stack, which layer runs the program?

- A. Drive motors
- B. V5 Brain
- C. Distance sensor
- D. Wheel tyres

Answer: B

Points: 1

8. What is differential steering on a Base Bot?

- A. Turning by running left and right sides at different speeds
- B. Using only one motor for all movement
- C. Steering with a car-style front axle only
- D. Turning only when the battery is low

Answer: A

Points: 1

9. Autonomous mode means:

- A. A student drives with the controller in real time
- B. The robot follows pre-programmed instructions without live driver input
- C. The robot cannot use sensors
- D. The Brain is switched off

Answer: B

Points: 1

10. A gear ratio that increases torque usually:

- A. Also increases top speed
- B. Reduces top speed on long runs
- C. Removes the need for a battery
- D. Disables encoder feedback

Answer: B

Points: 1

11. Before rewriting code for a skills match, which check comes first?

- A. Confirm autonomous and driver programs are downloaded and selected on the Brain
- B. Change motor power randomly
- C. Remove all sensors from the bot
- D. Delete the driver-control program

Answer: A

Points: 1

12. Low battery on a Base Bot often looks like:

- A. Stronger wireless range
- B. Weak or inconsistent motor drive
- C. Faster autonomous scoring
- D. Automatic sensor calibration

Answer: B

Points: 1

Short answer

13. In one or two sentences, explain why you should check motor cables before rewriting a drive program.

Open text

14. Name two sensors used on a Base Bot and state what each one measures.

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